BioH$_2$ production from food waste by anaerobic membrane bioreactor

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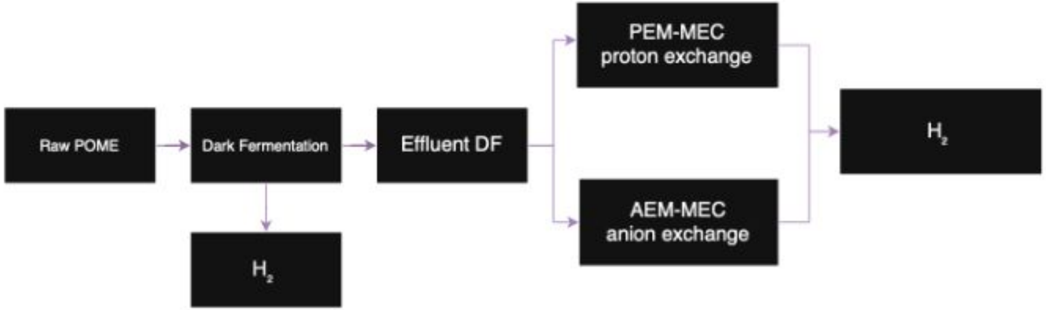


Introduction

The current

Global economy | Energy supply

depends largely on **fossil fuels**

Leading to

Hyper-consumption of
non-renewable resources

- Escalate **atmospheric CO2** concentration
- Depletion of **fossil fuel** resources

To **mitigate**: global warming & climate change

Fossil fuel	**Hydrogen**
• Climate change	• Generate H2O
• Non-biodegradable	• High energetic value **(120 MJ/kg)**
• Non-renewable	• High energy density **(143,35 kJ/kg)**

Process instability and **low hydrogen yields**

Hydrogen ***Production***

Couple of processes to produce hydrogen

- Biological processes
- Thermochemical processes

98%

76%	**22%**
Steam Methane forming	Coal Gasification

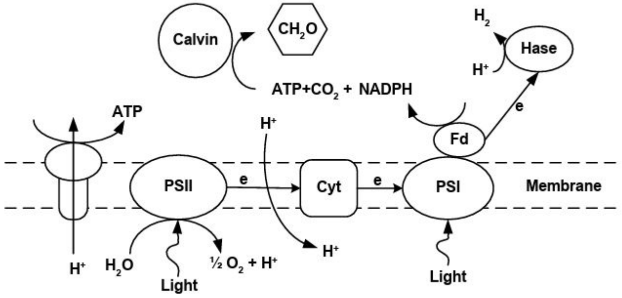
- Energy hungry
- Fossil fuel dependent
- 9 kg CO2/kg H2

Direct ***Biophotolysis***

Biohydrogen produced through **photosynthetic capability of an organism**

ex: green algae / cyanobacterium uses captured solar energy to drive water splitting process

- Producing O2
- Reducing ferredoxin
- Generating H2



- Low light conversion efficiencies
- Expensive hydrogen impermeable photobioreactors
- Oxygen-sensitive hydrogenase

Introduction

Anaerobic Membrane Bioreactors for Biohydrogen Production: Recent Developments, Challenges and Perspectives

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*Photo**fermentation***

An anaerobic photosynthesis by a **photosynthetic bacterium**

Organic acids

Acetic, lactic, butyric, malatic

solar energy | Anaerobic Sparse N

↓

H2 and CO2

Acetic acid

$$CH_3COOH + 2H_2O \rightarrow 4H_2 + 2CO_2$$

*Differences between **Light Dependent Processes***

Biophotolysis

- Splits H2O into O2, and makes H2

Photo**fermentation**

- Organic acids are oxidized to create H2

Light is used as solar energy (ATP) to drive the reaction.

*Dark **Fermentation***

- Not light dependent
  - Reactor design flexibility
  - Reactor configuration flexibility
  - Fermentative microorganism due to the absence of light:

Organic acids

Alcohols

valorized → **Energy** Source

- Requires dense substrates

***Food waste** (high strength substrate)*
- High production amount
- Widespread source
- Huge potential for valorization

1. Food waste has different bioH2 production potential
2. For commercial scale applications, High probability: composition of incoming food waste will have fluctuations

- **Carbohydrate-rich:** most suitable for biohydrogen production potential
- **Protein-rich and fat-rich:** possess less biohydrogen production potential
- **Cellulosic substances:** need to be broken down into smaller units before fermented

Previous Studies

1. Chu et al.
   "Two stage H2 and CH4 Production from Food Waste for Over 150 Days"
   HRT = 1.3 days
   pH = 5.5
2. Alpagani et al.
   "Biohydrogen production in CSTR for 251 Days"
   HRT = 5 days
   pH = 5.5
   Biohydrogen production
   104,3 mL/g VS added

Scope of Work

1. Evaluate effects of operating conditions of AnMBR to biohydrogen productio
2. Long term continuous system
3. Taking fluctuations in feedstock characterizations into account
4. To determine:
   - Optimum operating conditions
   - Valorizing microbial products: obtaining VFA

Anaerobic Digestion (AD)

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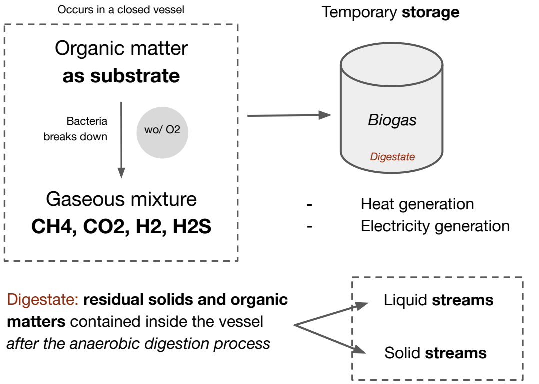
Natural biochemical process that

- Agricultural waste management
- Urban waste management

Recently,

Alternative energy **source**



fertilizer valuable plant nutrients

AD can process **a broad spectrum** of feedstock

Feedstock

- Removal of **non-biodegradable** components
  To: enhance biodegradablity
- Removal of **any toxic components**
  Prevent: impact bacteria negatively

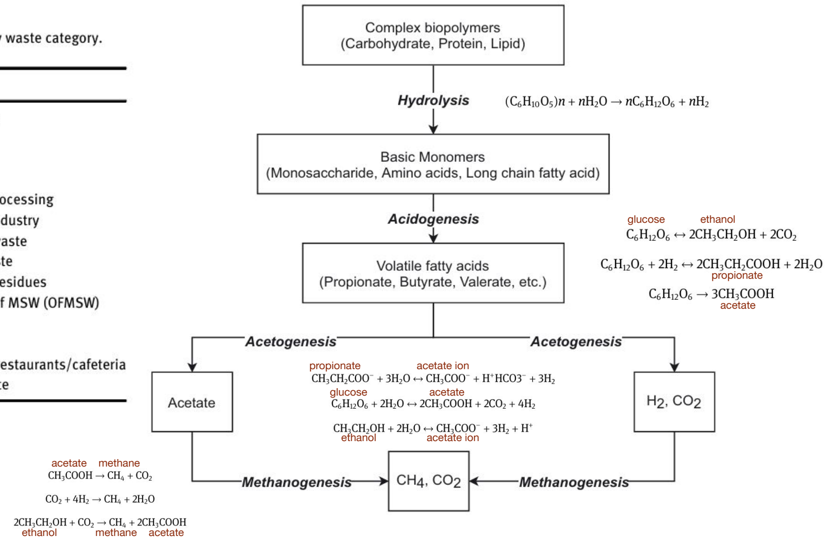
Table 1: Common anaerobic digestion (AD) feedstocks grouped by waste category.

Category	Source
Agricultural waste	Livestock manure
	Energy crops
	Harvest remains
	Farm mortality
Industrial waste	Food/beverage processing
	Pharmaceutical industry
	Slaughterhouse waste
	Dairy product waste
	Agro-processing residues
Municipal waste	Organic fraction of MSW (OFMSW)
	Sewage sludge
	Yard trimmings
	Food waste from restaurants/cafeteria
	Supermarket waste

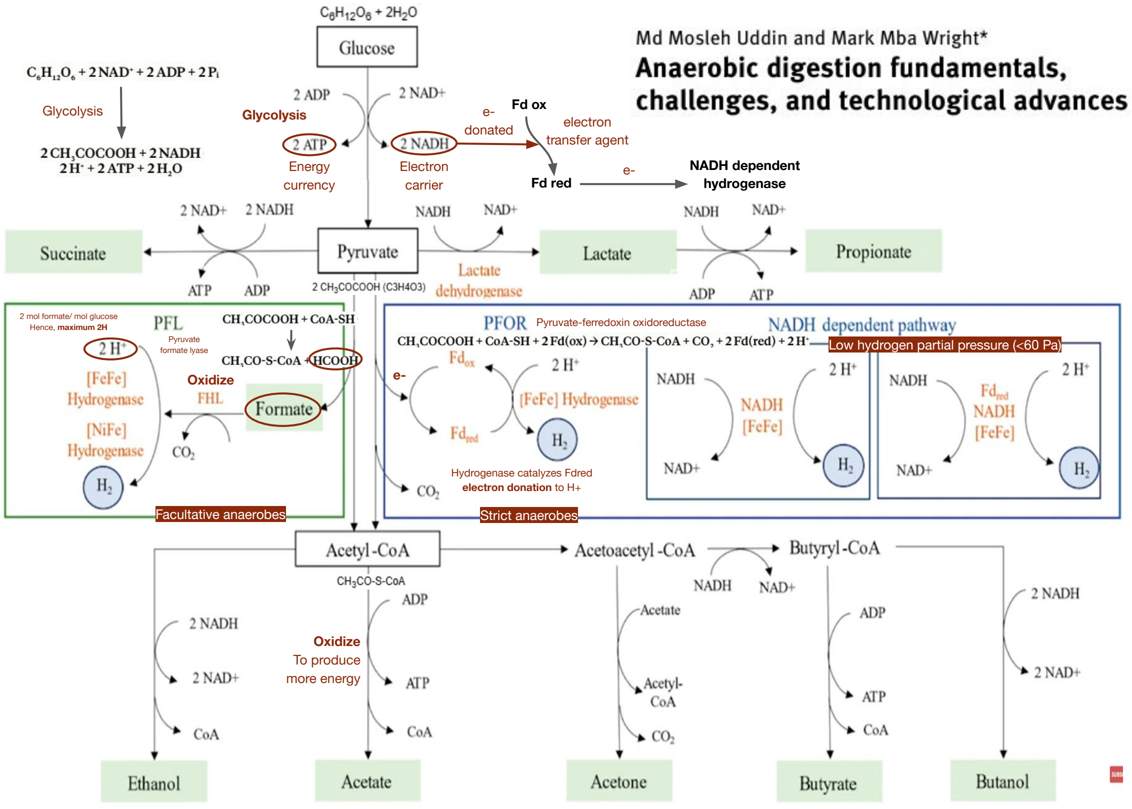
Biochemistry

AD occurs through multiple complex interaction between different **types of microorganisms**

Diverse microbial communities *collaborate*



Dark Fermentation



PFL vs PFOR Pathways

- PFL: facultative anaerobes, PFOR: strict
  1. **Facultative:** can grow with O2
  2. **Strict:** are killed by even a trace of O2
- PFL doesn't immediately **generate H2**
  1. **PFL:** needs FHL to activate hydrogenase
  2. **PFOR:** activates hydrogenase directly
- PFOR *theoretically* yields more biohydrogen from **1 glucose molecule**
  1. **PFL:** NADH -> reduced organic compounds
  2. **PFOR:** NADH -> 2 additional H

Dark fermentation alone **wastes most of its hydrogen potential** The leftover carbon leaves as **VFAs and alcohol** instead of H2

Anaerobic Membrane Bioreactor (AnMBR)

Membrane holds the biomass inside → while → **Letting treated liquid permeate**

- Decouples SRT from HRT
  1. **Short HRT:** for high throughput
  2. **Long SRT:** prevent slow-growing bacteria from washing out
  3. **Removes soluble VFAs and alcohols** that would inhibit fermentation

- Membrane fouling clogging of membrane by particles, biomass, and gunk.
  1. **Irreversible fouling:** particles smaller than the pores lodge inside the membrane *(hard to clean)*
  2. **Reversible fouling:** larger particles pile up on the surface

- Membrane materials
  1. **Polymeric membranes:** hydrophobic naturally (need surface modification)
  2. **Ceramic membranes:** resist heat, chemicals, corrosion, and fouling, expensive

AnMBR *Configurations*

- Submerged/immersed
  1. Membrane sits inside the reactor
  2. Liquid is sucked through it
  3. Trans-membrane pressure (TMP), a pressure difference, drives liquid across the membrane

Lower energy (no separate pumping loop)

- External/side-stream
  1. Mixed liquid is pumped out to **separate membrane unit** and **retained biomass returned**
  2. Higher TMP and velocity **(more energy use)**
  3. Easier to clean

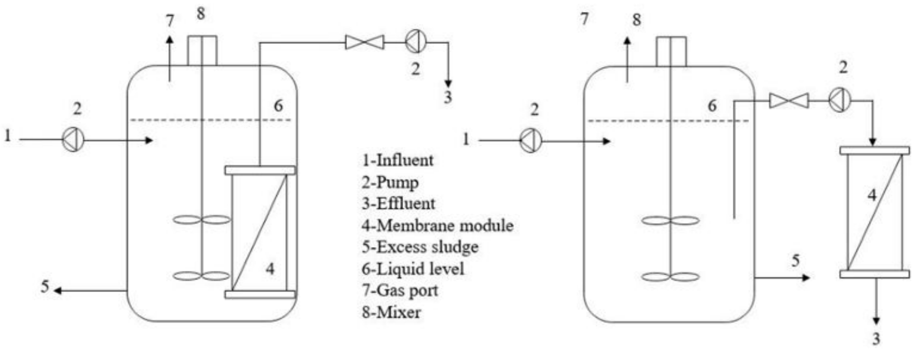


Fig. 2. Schematic representation of AnMBR (A) submerged, (B) external (Aslam et al., 2018).

Experiment

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Feedstock

- Foodwaste
  1. **Collected** every week
  2. From Istanbul Technical Uni **dining hall**
  3. **Stored @ 4 C** prior to daily feeding
- Characteristics
  1. **Different** regarding menu of the day
  2. Consisted of **food preparation leftovers** ex: peelings, rice, grains, pasta, veggies
- Pre-processing
  1. **Grinded** by **garbage disposer**
  2. **Screened** through a **sieve** (3 mm openings)
  3. All grounded food waste **passed through sieve** (95% of the time)

Inoculum

The seed of living microbes that carries out the fermentation to **kick start the process**

1. **Biomass** characterized MLSS (12.5 g/L), and
2. MLVSS (11 g/L)

Food waste @ 70 C, producing 65-80 mL H2/ g VS added

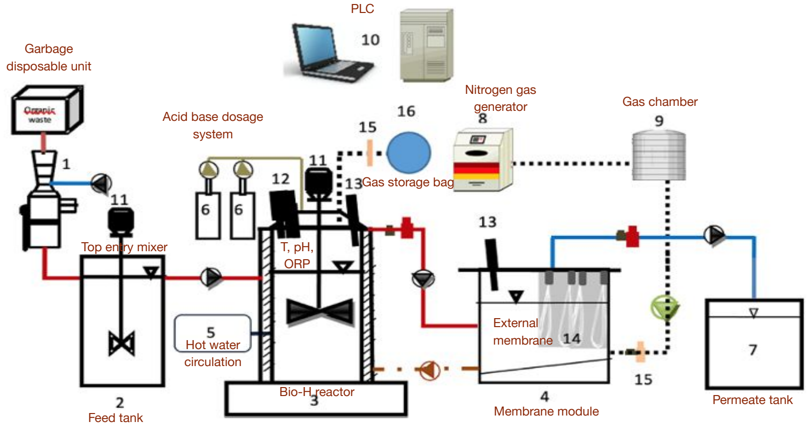
Reactor setup

- Continuously Stirred Tank Reactor (CSTR)
  - **Automated** laboratory scale reactor
  - V = 11,5 L, T = 55 C
  - N HCl = 2, N NaOH = 6, **to keep desired pH**
  - 868 days ; 8 periods

- Membrane module
  - **Submerged UF membrane module**
  - Pore size: 0,05 micro m

- Conditions given

P1 - P4: gradually increasing the OLR
From 4.8 to 27 g/L.day
P7-P8: low pH (5) to variable pH (5-8)



Microbial Electrolysis Cell (MEC) & Electrodialysis (ED)

Microbial Electrolysis Cell *(MEC)*

In the **anodic chamber** — Electrochemically active bacteria **oxidize the organic substrate** → into → **CO2 Protons Electrons**

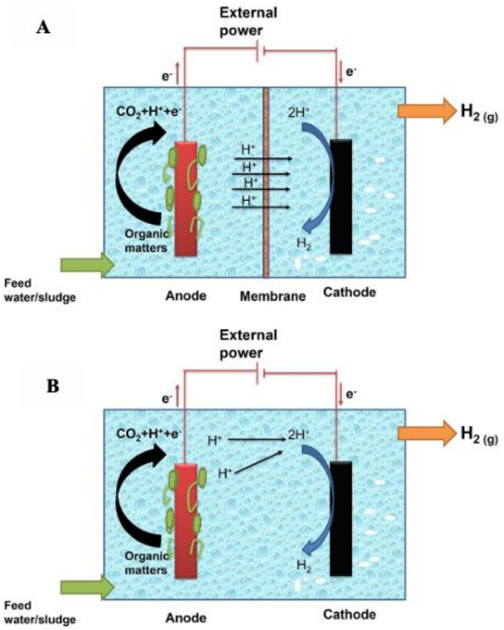
- Electrons travel through an external wire to the cathode

  Cathode: negative electrode where reduction happens
  -> Where **electrons meet protons** that have crossed PEM

  - Anode: $CH_3COO^- + 4H_2O \rightarrow 2HCO_3^- + 9H^+ + 8e^-$
  - Cathode: $8H^+ + 8e^- \rightarrow 4H_2$

- MEC needs an *externally applied voltage* (typically above 0.4 V)

- Anode and cathode are separated by ion-exchange membranes (IEM)
  1. IEM let specific ions through
  2. To keep H2 pure
  3. To stop anode bacteria from eating H



Electrodialysis *(ED)*

Removes **inhibitors** → so → **Fermentation makes more**

- Electrical field pulls ions across an **ion exchange membrane (IEM)**

Dilute stream → ions → Concentrated stream

- Applied to fermentation, continuously extracts the VFAs
  1. Prevents **VFA accumulation**
  2. Drop the **pH**
  3. Boosting **H2 yield**
  4. **Recovering the VFAs** as a product

Microbial Electrolysis Cell (MEC)

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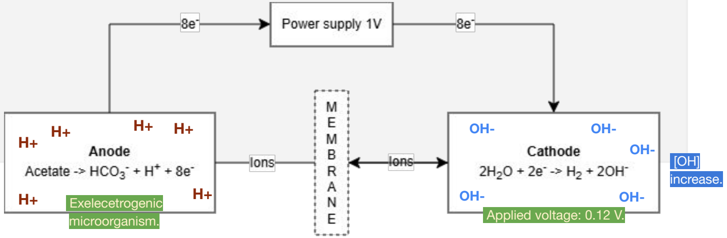
Microbial Electrolysis Cell (MEC)

Organic matter → convert → **Bio Hydrogen**

Waste water → degrade organic compounds → **Pure**



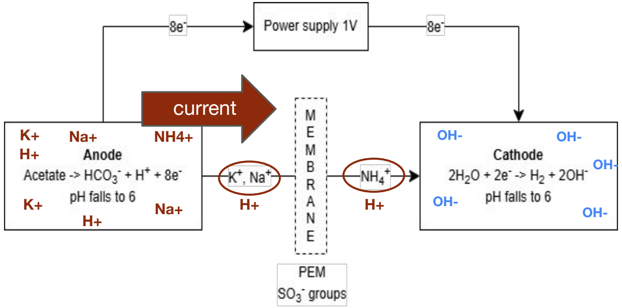
Anode chamber was continuously fed with synthetic wastewater.

- **NaCH3COO.3H2O,**0.74 g/L
- **KCl,** 0.58 g/L
- **NaCl,** 0.68 g /L
- **KH2PO4,** 0.87 g/L
- **K2HPO4,** 0.28 g/L
- **NH4Cl,** 0.1 g/L
- **MgSO4.7H2O,** 0.1 g/L
- **CaCl2.2H2O**

Proton Exchange Membrane

Membrane that has fixed negative groups, so it only passes cations (+).

H+, K+, Na+, NH4+, Mg2+.



Migrational contribution

depends on four things:

1. Charge
2. Mobility
3. Concentration
4. Electric field strength

Molar conductivity

H+ = 350 S·cm²/mol

Na+ = 50 S·cm²/mol

K+ = 74 S·cm²/mol

Cations instead are transported

1. Cation accumulates
2. Can't react with anions
3. Doesn't neutralize system
4. pH imbalance remains high

Concentration

[H+] = 10-7 M

[K+, Na+, NH4+, Mg2+] = 10-1 M

PEM vs AEM

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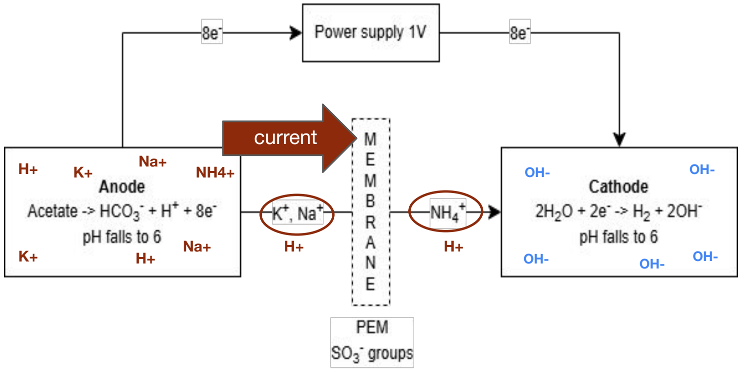
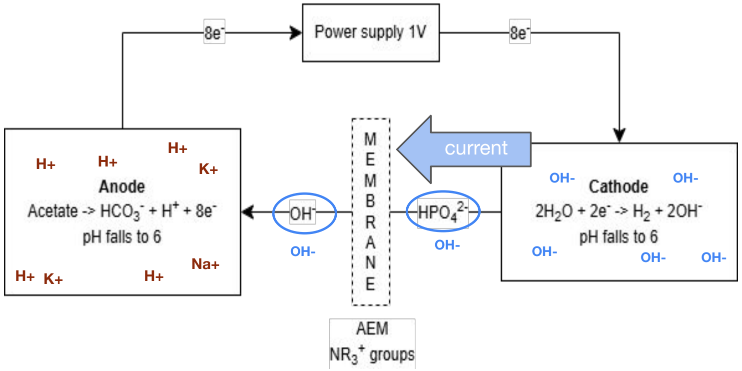
Proton Exchange Membrane
Membrane that has fixed negative groups, so it only passes cations (+).
H+ K+, Na+, NH4+, Mg2+.

Anion Exchange Membrane
positive groups, so it only passes anions (-).

OH-, HPO4-, PO4-, HCO3-,Cl-
Which all has a uniform concentration of 10-2 mol/L





Potential Losses

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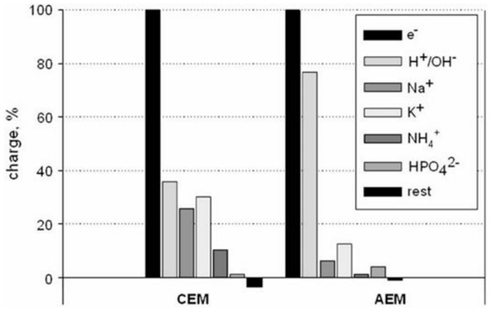
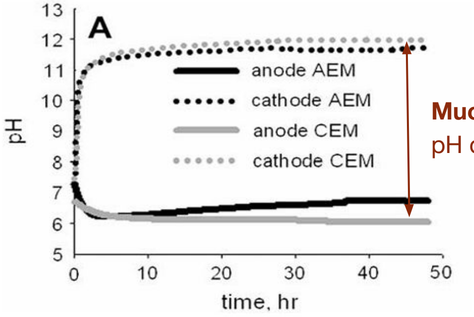


Fig. 6 – Comparison of the total charge production (e^-) to the transport of specific ions through CEM and AEM in an experimental run at an applied voltage of 1.0 V. Total charge production is set to 100% for relative comparison.



Much bigger,
pH difference

$$E_{eq} = E_{cat} - E_{an}$$

$$E_{eq} = \left(E^0_{cat} - \frac{RT}{2F} ln \frac{pH_2}{[10^{-7}]^2}\right) - \left(E^0_{an} - \frac{RT}{8F} ln \frac{[CH_3COO^-]}{[HCO_3{}^-]^2[10^{-7}]^9}\right)$$

The total losses

Applied voltage required for reduction reaction

The voltage we supplied

$$E_{cell} = E_{emf} + \boxed{E_{\Delta pH} + \eta_{an} + \eta_{cat} + E_{ionic} + E_T}$$

$$E_{cell} - E_{emf} = E_{\Delta pH} + \eta_{an} + \eta_{cat} + E_{ionic} + E_T$$

$$1.0V - 0.12V = E_{\Delta pH} + \eta_{an} + \eta_{cat} + E_{ionic} + E_T$$

pH unbalance

$$E_{\Delta pH} = \frac{RT}{F} \ln\left(10^{(pH_{cathode} - pH_{anode})}\right)$$

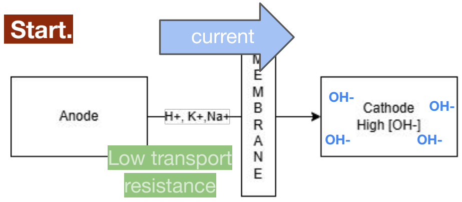
Overpotential, the extra voltage for reduction reaction that is dissipated as heat instead.

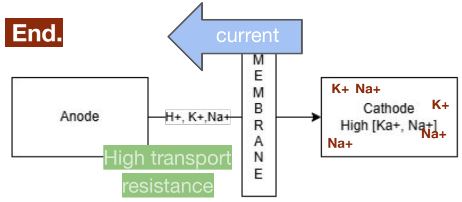
$$\eta_{an} = E_{an,measured} - E_{an}$$

$$\eta_{cat} = E_{cat} - E_{cat,measured}$$

Transport losses, the cost of voltage to drag ions across the membrane.

Caused by: accumulation of spectator cations
- Increasing acid concentration
- Switching the concentration gradient
- Increasing transport resistance





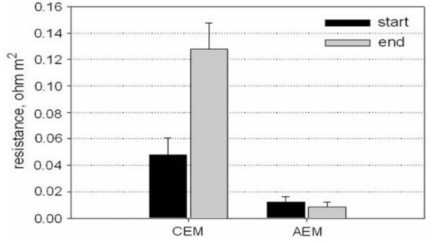


Fig. 4 – Transport resistance at the start and end of an experimental run in an MEC equipped with an AEM and in an MEC equipped with a CEM.

Potential Losses

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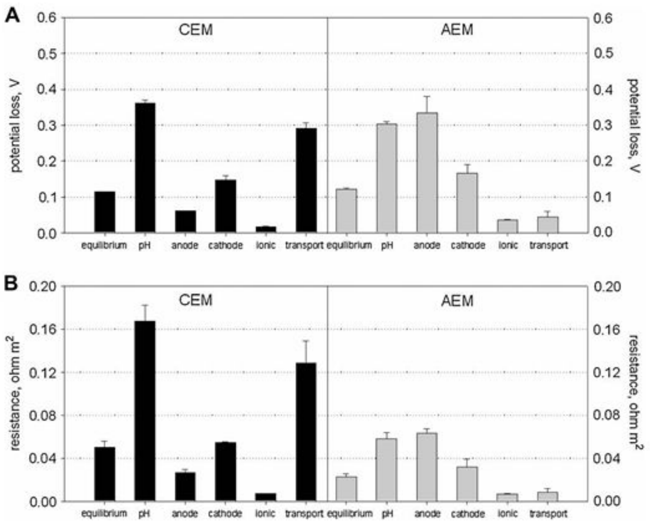

Table 1 – Comparison of the performance of two MECs equipped with a CEM and an AEM at an applied voltage of 1 V.

	Current density ($A\ m^{-2}$)	Cathodic efficiency (%)	Hydrogen production ($m^3\ H_2\ m^{-3}\ d^{-1}$)
CEM	2.3 ± 0.3	47 ± 6	0.4 ± 0.1
AEM	5.3 ± 0.5	83 ± 13	2.1 ± 0.5

$$E_{cell} = E_{emf} + E_{\Delta pH} + \eta_{an} + \eta_{cat} + E_{ionic} + E_T$$

$$I = \frac{V}{R} \Rightarrow j = \frac{E_{cell}}{R_{total}}$$

$$j_{CEM} = \frac{1V}{0.435\Omega m^2} = 2.3Am^{-2}$$

$$j_{AEM} = \frac{1V}{0.435\Omega m^2} = 5.3Am^{-2}$$

$$mol\ H_2/s = \frac{(j \cdot A \cdot \eta_{cat})}{2F}$$

- $A = electrode\ area = 0.025m^2$
- $F = 96485\ C/mol$
- $\eta_{cat} = cathodic\ efficiency$